

Step 7: Excel Analysis – Forecast Accuracy and Fulfilment KPIs

7.1 KPI Target:

KPI	Current (from Excel)	Target (PM assumption)	Gap
MAPE	3%	$\leq 10\%$	Exceeds Target
OTIF	25%	$\geq 95\%$	-70%
Fill Rate	98%	$\geq 97\%$	Meets Target
Stockout Rate	63%	$\leq 3\%$	+60%

MAPE, MAE, and Bias are calculated using forecast error logic based on the difference between Demand_Units and Forecast_Units, as documented in the analytical model and supporting evidence in Appendix C.

Note: OTIF in this project is calculated at period level using OTIF_Flag, where OTIF_Flag = 1 only when OnTime_Flag = 1 AND Stockout_Flag = 0 (on-time and in-full).

Targets represent indicative benchmarks used for analytical comparison and prioritisation rather than committed operational targets.

The contrast between strong forecast accuracy and weak fulfilment outcomes was the key analytical signal driving the remainder of the project.

7.2 KPI Interpretation:

Based on the KPI Summary derived from the Excel analysis:

- **Forecast accuracy is strong**, with a MAPE of **3%** and MAE of **48.4 units**, indicating that demand is being predicted reliably.
- **Forecast Bias of -10.1 units** suggests slight over-forecasting on average, but this is not material enough to cause major inefficiencies.

- **Fulfilment performance is weak**, with OTIF at only **25%**, despite an On-Time Delivery rate of **79%**.
- **Stockout Rate is very high (63%)**, indicating that in a majority of months, delivered quantities were insufficient to meet customer demand.
- **Fill Rate remains high at 98%**, implying that although shortages occur frequently, their magnitude is often small rather than severe.
- These KPI results form the basis of the project's operational diagnosis and are interpreted alongside benchmark fulfilment analysis rather than in isolation.

7.3 Root Cause Analysis:

Using the KPI results, the root causes can be summarised as follows:

Observation	Root Cause
Low OTIF (25%)	Combination of late deliveries and frequent stockouts
High Stockout Rate (63%)	Supplier under-delivery relative to demand
High Fill Rate (98%)	Shortfalls are usually marginal, not catastrophic
Good Forecast Accuracy	Demand planning capability is mature

7.4 Business Impact

From a business perspective, the current performance implies:

- **Customer dissatisfaction risk** due to frequent stockouts, even if quantities are small.
- **Revenue leakage**, as unmet demand cannot always be recovered.
- **Operational inefficiency**, with planners forecasting accurately but suppliers failing to execute.

- **Management blind spots**, as high fill rate may mask the true frequency of service failures.

This combination is particularly risky in a retail environment where **service reliability** drives repeat purchases.

7.5 Initial Improvement Opportunities (from KPI diagnosis):

Short-term (0–3 months) – Execution fixes

- Introduce exception triggers when Stockout_Flag = 1 or Fill Rate < 95%
- Conduct supplier performance reviews for months with late or short deliveries
- Implement weekly OTIF monitoring instead of monthly review

Medium-term (3–6 months) – Process improvements

- Introduce dynamic safety stock rules based on historical delivery reliability
- Segment suppliers by lead-time and reliability performance
- Align replenishment quantities more closely with execution risk, not just demand

Long-term (6–12 months) – Capability uplift

- Integrate AI-assisted exception forecasting to flag high-risk months
- Move from static replenishment to adaptive planning
- Embed KPIs into a Power BI operational dashboard for leadership visibility

7.6 Role Of AI :

While forecasting accuracy is already strong using traditional methods, AI-assisted analysis can be used in subsequent phases to identify exception patterns, supplier risk signals, and demand anomalies earlier. In this project, AI is positioned as a decision-support tool to enhance human judgement rather than replace planning logic.

7.7 Final Summary:

This demonstrated that while demand forecasting performance is strong, operational execution significantly limits service performance. The analysis highlights that improvements in supplier reliability, replenishment execution, and exception management will deliver greater business impact than further refinement of forecasting models alone.

Therefore, improving execution reliability will deliver higher ROI than further forecast optimisation.

STEP 9: KPI & Fulfilment Analysis (Excel + SQL + Dashboard-ready)

9.1 Purpose of KPI and Dashboard Analysis:

The purpose of this step is to analyse order fulfilment performance and demand behaviour using both operational KPIs (derived from Excel-based analysis) and large-scale benchmarking insights (derived from SQL analysis). The analysis aims to identify the root causes behind service-level failures, particularly low OTIF performance, and to present these insights through interactive dashboards designed for decision-makers.

Power BI is used as the primary visualisation tool to convert analytics-ready datasets into intuitive, executive-facing dashboards that support monitoring, investigation, and scenario-based decision-making.

The detailed operational KPI model is developed in the analytical layer of the project, while Power BI is used to visualise aggregated trends, benchmark fulfilment performance, and dashboard-based decision support for stakeholders.

9.2.1 KPI Summary Interpretation

The Excel-based KPI summary indicates mixed operational performance:

Forecast Accuracy (MAPE = 3%)

Forecast accuracy is strong, indicating that demand prediction is not a major source of operational risk.

These operational KPI findings are derived from the project’s simulated analytical model and should be interpreted separately from the external SQL benchmark metrics presented later in this section.

On-Time Delivery (79%)

A majority of orders are delivered on time; however, performance falls short of best-in-class benchmarks.

Stockout Rate (63%)

A high proportion of periods experience under-delivery relative to demand, signalling availability issues.

OTIF Performance (25%)

OTIF is significantly lower than both forecast accuracy and on-time delivery, indicating that service failures are driven by execution issues rather than demand planning.

OTIF Definition Used:

OTIF_Flag = 1 only when **OnTime_Flag = 1 AND Stockout_Flag = 0** (on-time and in-full).

9.2.2 Exception-Based Analysis

Exception periods were identified where either:

- Stockout_Flag = 1, or
- OnTime_Flag = 0

These exception months account for the majority of OTIF failures and highlight that even when deliveries are timely, under-delivery frequently prevents orders from being fulfilled in full.

This confirms that **supplier under-delivery and execution reliability**, rather than forecast error, are the dominant contributors to poor OTIF performance.

9.3 KPI Analysis – SQL Benchmark Dataset

9.3.1 SQL-Based Fulfilment KPIs

To provide external context, SQL was used to analyse a large-scale e-commerce dataset (Olist). The following benchmark KPIs were derived from the order_analytics view:

- **On-Time Delivery Percentage:** 90.45%
- **OTIF-like Performance:** 90.45%
- **Average Late Days:** 0.72 days

These results indicate that mature e-commerce operations can consistently achieve high delivery reliability with only marginal delivery delays.

9.3.2 Benchmark Interpretation

The SQL benchmark demonstrates that delivery timeliness can exceed 90% when operational execution is stable. Compared to the simulated operational dataset (On-Time 79%, OTIF 25%), the gap highlights that the primary improvement opportunity lies in execution reliability, particularly supplier fulfilment and delivery consistency.

The SQL dataset is used strictly as a **benchmarking and validation layer** and is not combined with the Excel KPIs.

9.4 Dashboard Development – Power BI

9.4.1 Dashboard Objectives

The Power BI dashboards were designed to:

- Provide executives with a high-level view of demand and fulfilment performance
- Enable operations teams to investigate service-level exceptions
- Support scenario analysis through interactive filters, targets, and bookmarks

9.4.2 Executive Overview Dashboard

The **Executive Overview** dashboard presents a consolidated view of performance using four headline KPIs:

- Total Orders
- Total Revenue
- Average Order Value

- On-Time Delivery Percentage

These measures are drawn from the aggregated monthly dashboard dataset and are intended to support executive visibility rather than replace the detailed analytical model.

Trend visuals show how order volumes, revenue, and on-time delivery evolve over time. A date slicer allows users to focus on specific periods and observe how service levels behave during demand peaks.

A Key Insights text panel summarises the most important observations, ensuring the dashboard supports storytelling rather than raw data inspection.

9.4.3 Operations & Exception Analysis

An operations-focused view enables deeper analysis of fulfilment behaviour:

- Trends in orders and revenue reveal demand seasonality
- On-time performance trends highlight periods of service degradation
- Scenario controls (Normal Demand vs Peak Demand) allow users to simulate operational stress conditions
- A what-if slider enables adjustment of target on-time performance to evaluate service gaps

These features support root-cause exploration and prioritisation of improvement actions.

9.4.4 Dashboard Acceptance Criteria:

- Cards must reconcile with the agreed KPI definitions and the underlying dashboard dataset.
- Slicer filters must affect all visuals consistently.
- Scenario buttons must change KPI view correctly.
- Insights box must update with scenario selection.

9.5 Interactive Features and Storytelling Elements

The dashboard incorporates several interactive elements to enhance decision support:

- **Slicers:** Enable time-based filtering
- **Bookmarks:** Allow switching between Normal Demand and Peak Demand scenarios
- **What-If Parameter:** Simulates changes in on-time performance targets
- **Dynamic Insight Text:** Updates contextually to guide interpretation

These features ensure the dashboard functions not only as a reporting tool but as an exploratory analytics interface.

9.6 Summary of Findings

The combined KPI and dashboard analysis leads to the following conclusions:

- Forecasting performance is strong and does not require replacement
- OTIF underperformance is primarily driven by execution failures rather than demand error
- Supplier under-delivery and delivery delays are the key operational risks
- Benchmark analysis confirms that significant improvement potential exists
- Interactive dashboards provide an effective mechanism for ongoing monitoring and exception management

9.7 Key Insights Summary

The analysis generated three core insights. First, forecast accuracy was relatively strong, indicating that demand prediction was not the primary performance constraint. Second, fulfilment outcomes remained weak, driven by frequent stockouts and inconsistent execution. Third, benchmark analysis showed that higher delivery reliability is achievable in practice, suggesting that the main improvement opportunity lies in operational execution rather than further forecast refinement.